LAPORAN PRAKTIKUM

MATA KULIAH:

PEMROGRAMAN MOBILE



Dosen Pengampu:

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TEKNIK INFORMATIKA

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Bagian A: Input Dasar dengan TextField

```dart
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const InputPage(),
    );
  }
}

class InputPage extends StatefulWidget {
  const InputPage({super.key});

  @override
  State<InputPage> createState() => _InputPageState();
}

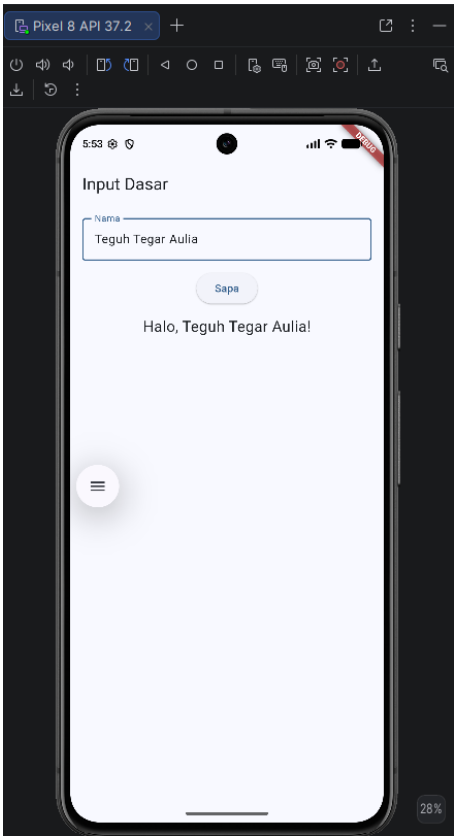
class _InputPageState extends State<InputPage> {
  final _controller = TextEditingController();
  String _salam = '';

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Input Dasar'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              decoration: const InputDecoration(
                labelText: 'Nama',
                border: OutlineInputBorder(),
              ),
            ),
            const SizedBox(height: 12),
            ElevatedButton(
              onPressed: () {
                setState(() => _salam = 'Halo, ${_controller.text}!');
              },
```

```
              child: const Text('Sapa'),
            ),
            const SizedBox(height: 12),
            Text(
              _salam,
              style: const TextStyle(fontSize: 20),
            ),
          ],
        ),
      ),
    );
  }
}
```

Hasil :



Bagian B: Form dengan Validasi

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(
```

```
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const FormPage(),
    );
  }
}

class FormPage extends StatefulWidget {
  const FormPage({super.key});

  @override
  State<FormPage> createState() => _FormPageState();
}

class _FormPageState extends State<FormPage> {
  final _formKey = GlobalKey<FormState>();
  final _nama = TextEditingController();
  final _email = TextEditingController();
  String? _jurusan;
  bool _setuju = false;

  @override
  void dispose() {
    _nama.dispose();
    _email.dispose();
    super.dispose();
  }

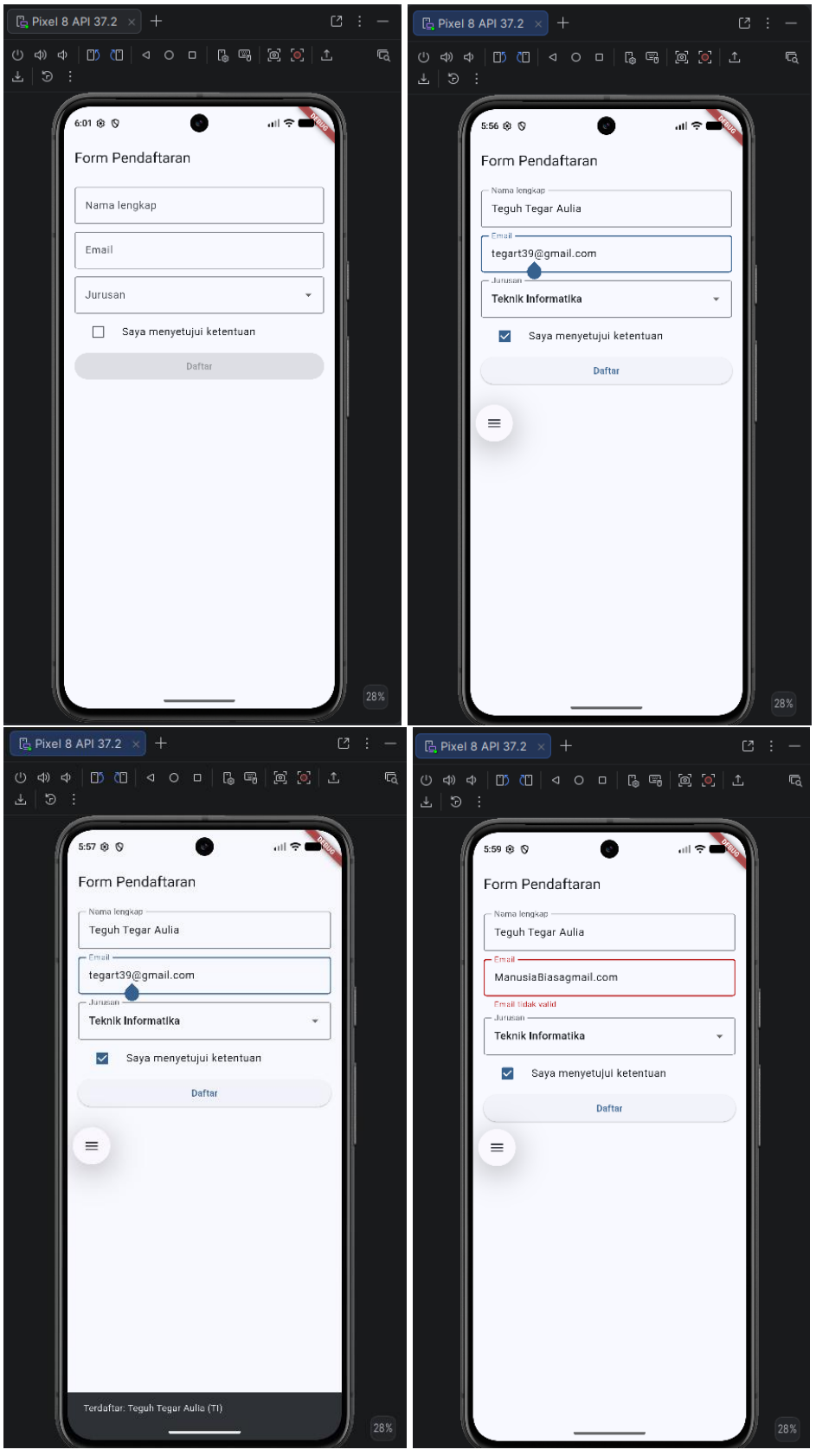
  void _kirim() {
    if (_formKey.currentState!.validate()) {
      final jurusan = _jurusan ?? '-';

      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(
          content: Text('Terdaftar: ${_nama.text} ($jurusan)'),
        ),
      );
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Form Pendaftaran'),
      ),
      body: Form(
        key: _formKey,
        child: ListView(
          padding: const EdgeInsets.all(16),
          children: [
            TextFormField(
              controller: _nama,
              decoration: const InputDecoration(
                labelText: 'Nama lengkap',
                border: OutlineInputBorder(),
              ),
              validator: (v) =>
              (v == null || v.trim().isEmpty)
                  ? 'Nama wajib diisi'
```

```
                : null,
          ),
          const SizedBox(height: 12),
          TextFormField(
            controller: _email,
            keyboardType: TextInputType.emailAddress,
            decoration: const InputDecoration(
              labelText: 'Email',
              border: OutlineInputBorder(),
            ),
            validator: (v) {
              if (v == null || !v.contains('@')) {
                return 'Email tidak valid';
              }
              return null;
            },
          ),
          const SizedBox(height: 12),
          DropdownButtonFormField<String>(
            decoration: const InputDecoration(
              labelText: 'Jurusan',
              border: OutlineInputBorder(),
            ),
            items: const [
              DropdownMenuItem(
                value: 'TI',
                child: Text('Teknik Informatika'),
              ),
              DropdownMenuItem(
                value: 'SI',
                child: Text('Sistem Informasi'),
              ),
              DropdownMenuItem(
                value: 'TE',
                child: Text('Teknik Elektro'),
              ),
            ],
            onChanged: (v) => setState(() => _jurusan = v),
            validator: (v) => v == null ? 'Pilih jurusan' : null,
          ),
          CheckboxListTile(
            title: const Text('Saya menyetujui ketentuan'),
            value: _setuju,
            controlAffinity: ListTileControlAffinity.leading,
            onChanged: (v) => setState(() => _setuju = v ?? false),
          ),
          ElevatedButton(
            onPressed: _setuju ? _kirim : null,
            child: const Text('Daftar'),
          ),
        ],
      ),
    ),
  );
 }
}
```

Hasil :


Pixel 8 API 37.2
Form Pendaftaran
Nama lengkap
Email
Jurusan
Saya menyetujui ketentuan
Daftar
Teguh Tegar Aulia
tegart39@gmail.com
Teknik Informatika
ManusiaBiasagmail.com
Email tidak valid
Terdaftar: Teguh Tegar Aulia (TI)

Bagian D: Aplikasi Daftar Tugas dengan Provider

```dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;

  Tugas(this.judul, {this.selesai = false});
}

class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];

  List<Tugas> get items => List.unmodifiable(_items);

  int get jumlahSelesai => _items.where((t) => t.selesai).length;

  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const TugasPage(),
    );
  }
}

class TugasPage extends StatelessWidget {
```

```dart
  const TugasPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text(
          'Tugas (${model.jumlahSelesai}/${model.items.length})',
        ),
      ),
      body: ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];

          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) {
                context.read<TugasModel>().toggle(i);
              },
            ),
            title: Text(
              t.judul,
              style: TextStyle(
                decoration: t.selesai
                    ? TextDecoration.lineThrough
                    : null,
              ),
            ),
            trailing: IconButton(
              icon: const Icon(Icons.delete),
              onPressed: () {
                context.read<TugasModel>().hapus(i);
              },
            ),
          );
        },
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.push(
            context,
            MaterialPageRoute(
              builder: (_) => const TambahPage(),
            ),
          );
        },
        child: const Icon(Icons.add),
      ),
    );
  }
}

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
```

```dart
}

class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

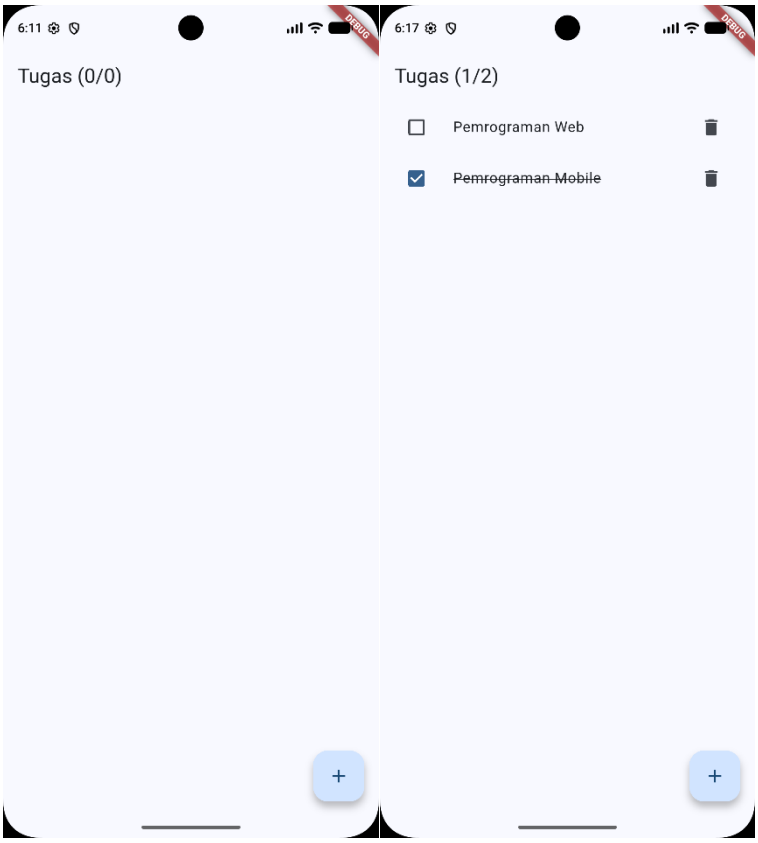
  void _simpan() {
    final judul = _controller.text.trim();

    if (judul.isEmpty) return;

    context.read<TugasModel>().tambah(judul);
    Navigator.pop(context);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Tambah Tugas'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              autofocus: true,
              decoration: const InputDecoration(
                labelText: 'Judul tugas',
                border: OutlineInputBorder(),
              ),
              onSubmitted: (_) => _simpan(),
            ),
            const SizedBox(height: 12),
            ElevatedButton(
              onPressed: _simpan,
              child: const Text('Simpan'),
            ),
          ],
        ),
      ),
    );
  }
}
```

Hasil :



Latihan Mandiri

1. Tambahkan validasi pada TambahPage: judul minimal 3 karakter dan tampilkan pesan error (ubah ke TextFormField + Form).
2. Tambahkan method hapusSelesai() pada TugasModel dan tombol di AppBar untuk menghapus semua tugas yang sudah selesai.
3. Tampilkan SnackBar "Tugas ditambahkan" setelah tugas disimpan.
4. Bila daftar kosong, tampilkan teks "Belum ada tugas" di tengah layar.

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;

  Tugas(this.judul, {this.selesai = false});
}

class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];

  List<Tugas> get items => List.unmodifiable(_items);

  int get jumlahSelesai => _items.where((t) => t.selesai).length;
```

```dart
  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }

  void hapusSelesai() {
    _items.removeWhere((t) => t.selesai);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const TugasPage(),
    );
  }
}

class TugasPage extends StatelessWidget {
  const TugasPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text(
          'Tugas (${model.jumlahSelesai}/${model.items.length})',
        ),
        actions: [
          IconButton(
            icon: const Icon(Icons.delete),
```

```
            onPressed: () {
              context.read<TugasModel>().hapusSelesai();
            },
          ),
        ],
      ),
      body: model.items.isEmpty
          ? const Center(
        child: Text('Belum ada tugas'),
      )
          : ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];

          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) {
                context.read<TugasModel>().toggle(i);
              },
            ),
            title: Text(
              t.judul,
              style: TextStyle(
                decoration: t.selesai
                    ? TextDecoration.lineThrough
                    : null,
              ),
            ),
            trailing: IconButton(
              icon: const Icon(Icons.delete),
              onPressed: () {
                context.read<TugasModel>().hapus(i);
              },
            ),
          );
        },
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.push(
            context,
            MaterialPageRoute(
              builder: (_) => const TambahPage(),
            ),
          );
        },
        child: const Icon(Icons.add),
      ),
    );
  }
}

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
}
```

```dart
class _TambahPageState extends State<TambahPage> {
  final _formKey = GlobalKey<FormState>();
  final _controller = TextEditingController();

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

  void _simpan() {
    if (_formKey.currentState!.validate()) {
      final judul = _controller.text.trim();

      context.read<TugasModel>().tambah(judul);

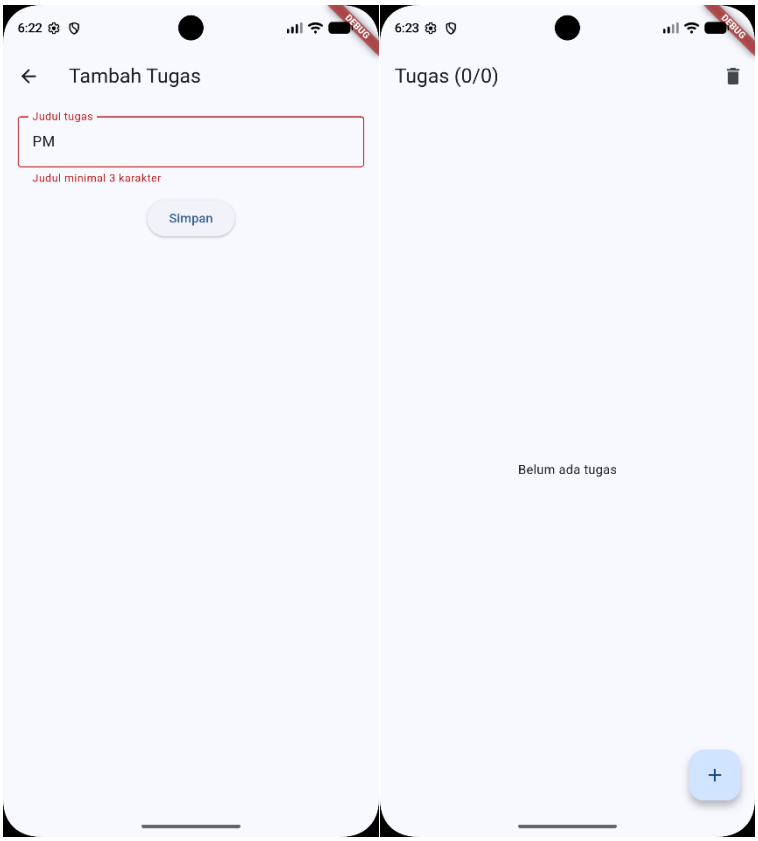
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text('Tugas ditambahkan'),
        ),
      );

      Navigator.pop(context);
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Tambah Tugas'),
      ),
      body: Form(
        key: _formKey,
        child: Padding(
          padding: const EdgeInsets.all(16),
          child: Column(
            children: [
              TextFormField(
                controller: _controller,
                autofocus: true,
                decoration: const InputDecoration(
                  labelText: 'Judul tugas',
                  border: OutlineInputBorder(),
                ),
                validator: (v) {
                  if (v == null || v.trim().length < 3) {
                    return 'Judul minimal 3 karakter';
                  }
                  return null;
                },
                onFieldSubmitted: (_) => _simpan(),
              ),
              const SizedBox(height: 12),
              ElevatedButton(
                onPressed: _simpan,
                child: const Text('Simpan'),
              ),
            ],
          ),
        ),
```

```
      ),
    );
  }
}
```

Hasil :



Tugas aplikasi Daftar Belanja

1. Halaman Form Tambah: nama barang (wajib), jumlah (wajib, angka lebih dari 0), dan kategori (dropdown), dengan validasi pada setiap isian.
2. Halaman Daftar: menampilkan semua barang, dapat dicentang sebagai "sudah dibeli" dan dihapus.
3. State disimpan pada satu ChangeNotifier dan dibagikan dengan Provider; AppBar menampilkan jumlah barang yang belum dibeli.
4. Kumpulkan: tangkapan layar kedua halaman (termasuk tampilan pesan error validasi) dan berkas main.dart (atau tautan repositori).

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Barang {
  String nama, kategori;
  int jumlah;
  bool dibeli;
```

```
  Barang(this.nama, this.jumlah, this.kategori, {this.dibeli = false});
}

class BelanjaModel extends ChangeNotifier {
  final List<Barang> _items = [];

  List<Barang> get items => _items;
  int get belumDibeli => _items.where((b) => !b.dibeli).length;

  void tambah(String nama, int jumlah, String kategori) {
    _items.add(Barang(nama, jumlah, kategori));
    notifyListeners();
  }

  void toggle(int i) {
    _items[i].dibeli = !_items[i].dibeli;
    notifyListeners();
  }

  void hapus(int i) {
    _items.removeAt(i);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => BelanjaModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Belanja',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const BelanjaPage(),
    );
  }
}

class BelanjaPage extends StatelessWidget {
  const BelanjaPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<BelanjaModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text('Daftar Belanja (${model.belumDibeli})'),
```

```
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),
      body: model.items.isEmpty
          ? const Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              Icons.shopping_cart_outlined,
              size: 80,
              color: Colors.blue,
            ),
            SizedBox(height: 16),
            Text(
              'Belum ada barang',
              style: TextStyle(fontSize: 20),
            ),
          ],
        ),
      )
          : ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final b = model.items[i];

          return Card(
            margin: const EdgeInsets.symmetric(
              horizontal: 12,
              vertical: 6,
            ),
            child: ListTile(
              leading: CircleAvatar(
                backgroundColor: Colors.blue.shade100,
                child: Icon(
                  Icons.shopping_bag,
                  color: Colors.blue,
                ),
              ),
              title: Text(
                b.nama,
                style: TextStyle(
                  fontSize: 18,
                  fontWeight: FontWeight.bold,
                  decoration: b.dibeli
                      ? TextDecoration.lineThrough
                      : null,
                ),
              ),
              subtitle: Text(
                '${b.jumlah} • ${b.kategori}',
              ),
              trailing: Row(
                mainAxisSize: MainAxisSize.min,
                children: [
                  Checkbox(
                    value: b.dibeli,
                    onChanged: (_) => model.toggle(i),
                  ),
                  IconButton(
                    icon: const Icon(
```

```
                    Icons.delete,
                    color: Colors.red,
                  ),
                  onPressed: () => model.hapus(i),
                ),
              ],
            ),
          ),
        );
      },
    ),
    floatingActionButton: FloatingActionButton(
      onPressed: () {
        Navigator.push(
          context,
          MaterialPageRoute(
            builder: (_) => const TambahPage(),
          ),
        );
      },
      backgroundColor: Colors.blue,
      foregroundColor: Colors.white,
      child: const Icon(Icons.add),
    ),
  );
 }
}

class TambahPage extends StatefulWidget {
 const TambahPage({super.key});

 @override
 State<TambahPage> createState() => _TambahPageState();
}

class _TambahPageState extends State<TambahPage> {
 final _formKey = GlobalKey<FormState>();
 final _nama = TextEditingController();
 final _jumlah = TextEditingController();
 String? _kategori;

 @override
 void dispose() {
  _nama.dispose();
  _jumlah.dispose();
  super.dispose();
 }

 void _simpan() {
  if (!_formKey.currentState!.validate()) return;

  context.read<BelanjaModel>().tambah(
    _nama.text,
    int.parse(_jumlah.text),
    _kategori!,
  );

  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text('Barang ditambahkan'),
    ),
```

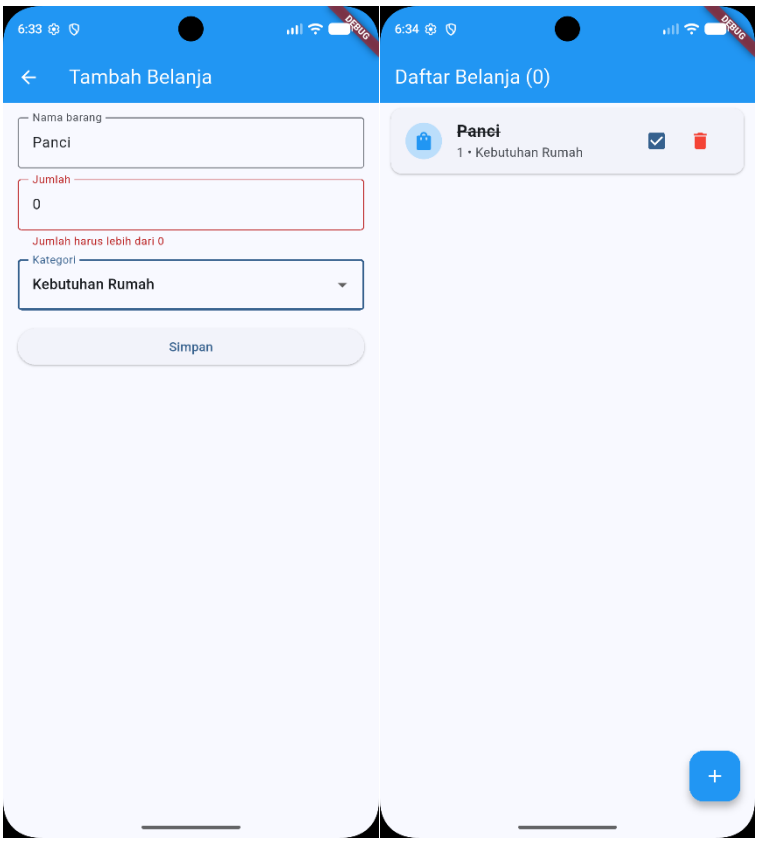
```
    );

    Navigator.pop(context);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Tambah Belanja'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),
      body: Form(
        key: _formKey,
        child: Padding(
          padding: const EdgeInsets.all(16),
          child: Column(
            children: [
              TextFormField(
                controller: _nama,
                decoration: const InputDecoration(
                  labelText: 'Nama barang',
                  border: OutlineInputBorder(),
                ),
                validator: (v) =>
                v == null || v.trim().isEmpty
                    ? 'Nama wajib diisi'
                    : null,
              ),
              const SizedBox(height: 12),
              TextFormField(
                controller: _jumlah,
                keyboardType: TextInputType.number,
                decoration: const InputDecoration(
                  labelText: 'Jumlah',
                  border: OutlineInputBorder(),
                ),
                validator: (v) {
                  final jumlah = int.tryParse(v ?? '');
                  return jumlah == null || jumlah <= 0
                      ? 'Jumlah harus lebih dari 0'
                      : null;
                },
              ),
              const SizedBox(height: 12),
              DropdownButtonFormField<String>(
                decoration: const InputDecoration(
                  labelText: 'Kategori',
                  border: OutlineInputBorder(),
                ),
                items: const [
                  DropdownMenuItem(
                    value: 'Makanan',
                    child: Text('Makanan'),
                  ),
                  DropdownMenuItem(
                    value: 'Minuman',
                    child: Text('Minuman'),
                  ),
                  DropdownMenuItem(
```

```
                    value: 'Kebutuhan Rumah',
                    child: Text('Kebutuhan Rumah'),
                  ),
                  DropdownMenuItem(
                    value: 'Lainnya',
                    child: Text('Lainnya'),
                  ),
                ],
                onChanged: (v) => setState(() => _kategori = v),
                validator: (v) =>
                v == null ? 'Pilih kategori' : null,
              ),
              const SizedBox(height: 16),
              SizedBox(
                width: double.infinity,
                child: ElevatedButton(
                  onPressed: _simpan,
                  child: const Text('Simpan'),
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

Hasil :



Pertanyaan Refleksi

1. Mengapa TextEditingController harus di-dispose?
2. Kapan cukup memakai setState, dan kapan sebaiknya beralih ke Provider?
3. Apa yang terjadi bila notifyListeners() lupa dipanggil? Mengapa?
4. Mengapa di dalam onPressed kita memakai context.read, bukan context.watch?

Jawaban

1. TextEditingController harus di dispose supaya resource yang dipakai controller dilepas dan tidak menyebabkan kebocoran memori.
2. **setState** cukup untuk state sederhana dalam satu halaman. Provider lebih cocok kalau state perlu dipakai atau dibagikan ke beberapa halaman/widget.
3. Kalau notifyListeners() lupa dipanggil, data memang berubah, tetapi widget yang memakai watch tidak tahu kalau ada perubahan, jadi tampilan tidak otomatis diperbarui.
4. Di onPressed memakai context.read karena kita hanya perlu menjalankan perubahan data, bukan mendengarkan perubahan tersebut. watch digunakan ketika widget perlu ikut diperbarui saat data berubah.